
Preface

When ChatGPT came out, like many of my colleagues, I was disoriented. What surprised me wasn't the model's size or capabilities. For over a decade, the AI community has known that scaling up a model improves it. In 2012, the AlexNet authors noted in **their landmark paper** that: "All of our experiments suggest that our results can be improved simply by waiting for faster GPUs and bigger datasets to become available."^{1, 2}

What surprised me was the sheer number of applications this capability boost unlocked. I thought a small increase in model quality metrics might result in a modest increase in applications. Instead, it resulted in an explosion of new possibilities.

Not only have these new AI capabilities increased the demand for AI applications, but they have also lowered the entry barrier for developers. It's become so easy to get started with building AI applications. It's even possible to build an application without writing a single line of code. This shift has transformed AI from a specialized discipline into a powerful development tool everyone can use.

Even though AI adoption today seems new, it's built upon techniques that have been around for a while. Papers about language modeling came out as early as the 1950s. Retrieval-augmented generation (RAG) applications are built upon retrieval technology that has powered search and recommender systems since long before the term RAG was coined. The best practices for deploying traditional machine learning applications—systematic experimentation, rigorous evaluation, relentless optimization for faster and cheaper models—are still the best practices for working with foundation model-based applications.

1 An author of the AlexNet paper, Ilya Sutskever, went on to cofound OpenAI, turning this lesson into reality with GPT models.

2 Even **my small project in 2017**, which used a language model to evaluate translation quality, concluded that we needed "a better language model."

The familiarity and ease of use of many AI engineering techniques can mislead people into thinking there is nothing new to AI engineering. But while many principles for building AI applications remain the same, the scale and improved capabilities of AI models introduce opportunities and challenges that require new solutions.

This book covers the end-to-end process of adapting foundation models to solve real-world problems, encompassing tried-and-true techniques from other engineering fields and techniques emerging with foundation models.

I set out to write the book because I wanted to learn, and I did learn a lot. I learned from the projects I worked on, the papers I read, and the people I interviewed. During the process of writing this book, I used notes from over 100 conversations and interviews, including researchers from major AI labs (OpenAI, Google, Anthropic, ...), framework developers (NVIDIA, Meta, Hugging Face, Anyscale, LangChain, LlamaIndex, ...), executives and heads of AI/data at companies of different sizes, product managers, community researchers, and independent application developers (see “[Acknowledgments](#)” on page xx).

I especially learned from early readers who tested my assumptions, introduced me to different perspectives, and exposed me to new problems and approaches. Some sections of the book have also received thousands of comments from the community after being shared on [my blog](#), many giving me new perspectives or confirming a hypothesis.

I hope that this learning process will continue for me now that the book is in your hands, as you have experiences and perspectives that are unique to you. Please feel free to share any feedback you might have for this book with me via [X](#), [LinkedIn](#), or email at hi@huyenchip.com.

What This Book Is About

This book provides a framework for adapting foundation models, which include both large language models (LLMs) and large multimodal models (LMMs), to specific applications.

There are many different ways to build an application. This book outlines various solutions and also raises questions you can ask to evaluate the best solution for your needs. Some of the many questions that this book can help you answer are:

- Should I build this AI application?
- How do I evaluate my application? Can I use AI to evaluate AI outputs?
- What causes hallucinations? How do I detect and mitigate hallucinations?
- What are the best practices for prompt engineering?
- Why does RAG work? What are the strategies for doing RAG?

- What’s an agent? How do I build and evaluate an agent?
- When to finetune a model? When not to finetune a model?
- How much data do I need? How do I validate the quality of my data?
- How do I make my model faster, cheaper, and secure?
- How do I create a feedback loop to improve my application continually?

The book will also help you navigate the overwhelming AI landscape: types of models, evaluation benchmarks, and a seemingly infinite number of use cases and application patterns.

The content in this book is illustrated using case studies, many of which I worked on, backed by ample references and extensively reviewed by experts from a wide range of backgrounds. Although the book took two years to write, it draws from my experience working with language models and ML systems from the last decade.

Like my previous O’Reilly book, *Designing Machine Learning Systems* (DMLS), this book focuses on the fundamentals of AI engineering instead of any specific tool or API. Tools become outdated quickly, but fundamentals should last longer.³

Reading *AI Engineering* (AIE) with *Designing Machine Learning Systems* (DMLS)

AIE can be a companion to DMLS. DMLS focuses on building applications on top of traditional ML models, which involves more tabular data annotations, feature engineering, and model training. AIE focuses on building applications on top of foundation models, which involves more prompt engineering, context construction, and parameter-efficient finetuning. Both books are self-contained and modular, so you can read either book independently.

Since foundation models are ML models, some concepts are relevant to working with both. If a topic is relevant to AIE but has been discussed extensively in DMLS, it’ll still be covered in this book, but to a lesser extent, with pointers to relevant resources.

Note that many topics are covered in DMLS but not in AIE, and vice versa. The first chapter of this book also covers the differences between traditional ML engineering and AI engineering. A real-world system often involves both traditional ML models and foundation models, so knowledge about working with both is often necessary.

³ Teaching a course on how to use TensorFlow in 2017 taught me a painful lesson about how quickly tools and tutorials become outdated.